

Geological Containment Reliability

Definition study for development of an analysis tool (CORA)

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1001364-002

Title
Geological Containment Reliability Analysis

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1 Introduction

1.1 Contractual references

This report describes the result of a definition study on a tool for reliability assessment of geological containment storage. The reported work by Stichting Deltares (SD) has been executed on behalf of Stichting Shell research, under Shell contract number SSR 152. The current project leader at Shell side is Dr. P. van den Bogert.

The considered tool was in the contract provisionally called *Reliacon* (**RELI**Ability analysis of **CON**tainments). The finally adopted name is *CORA* (**CON**tainment **RE**liability **AS**essment).

1.2 Summary of the project stages, according to the project plan

The project plan attached to the contract describes five stages, to be executed partly by Shell employees and partly by Deltares employees. The original stage descriptions are summarized below.

1. Kick-off and definition of the required output of the framework.
This includes an analysis of the context, in which the framework is being developed, and the intended use. It has to result in the definition of the required output of the calculation framework and concerns the type and form of information as well as the required degree of precision. Another aspect to be defined is the degree of user-friendliness that is aimed for. This can impose limitations on the types of models to be used, especially in terms of calculation time.
2. Identification of failure modes and system definition.
An inventory of the potential hazards for containment storage is made. These are defined in terms of failure modes. The failure modes will be combined in a fault tree, which represents the containment system in terms of reliability analysis. Attention has to be paid to possible (strong) correlations between different failure modes based on physical reasons or whatsoever.
3. Inventory of the models applied per failure mode (mainly Shell task, evaluation and conclusions by SD).
For the identified failure modes an inventory of the applied models in terms of safety assessment has to be made. This inventory will allow us to concretize the analytical system description and to define the scope of the framework. Furthermore, it will enable us to select or prioritize models for the prototype. The description should include: the type of the model; input and output; availability; model uncertainty; application areas, application conditions and application limitations; comparison with alternative models.
4. Inventory and analysis of monitoring activities related to the failure modes (mainly Shell task, evaluation and conclusions by SD)
The final system is expected to support decisions on preventive measures for risk reduction on a cost basis. For this purpose it has to be analyzed per failure mode which kind of monitoring is applicable to prevent and control the failure occurrence.

Furthermore, it has to be analyzed how the monitoring or surveillance data affects the reliability assessment. At this stage, a decision on the strategy for further implementation of monitoring data should be made. This issue could be worked out in parallel to the development of the framework, or alternatively the focus would be on the framework first and the integration of monitoring as an extension would be carried out subsequently.

5. Definition study / DECISION POINT 1 / Quotation phases 6 to 13

The outcomes of stages 1 to 4 are elaborated in form of a definition study for the theoretical framework and the software to be developed. The results will be summarized in a report, based on which the setup of the subsequent phases will be defined. According to the agreed setup a quotation for phases 6 to 13 will be prepared. The scope of work will be process respectively capacity driven, i.e. the items will be developed in order of priority. In the quotation the content of the framework and the specifications for the software prototype will be described, including the specifications for verification and testing, i.e. the acceptance criteria for the prototype. Also subcontracting portions of the work should be discussed prior to making the quotation, based on the expected effort and benefits arising from the definition study, as well as the question of software ownership, maintenance aspects etc.

1.3 Project deliverable, according to the contract

The contract states that the deliverable is a report, consisting of the following two parts:

- a. Results of phases 1 to 4, containing the inventories of the considered mechanisms, models, uncertainties, monitoring activities and a description of the system in terms of a fault tree.
- b. Specifications of the theoretical framework and the requirements for the input, output and functionality of the software tool.

1.4 Readers' Guide

To be added

2 Inventory and scoping

A rough inventory of relevant aspects, including choices for the scope is presented hereafter. The inventory includes a number of pictures and tables from background literature. These pictures and tables are sometimes presented without a reference to the original source.

2.1 Geological storage

Containment storage to be considered is not only CO₂, but also storage of other liquids and gasses.

CORA will support analysis of the following geological storage types:

- Depleted hydrocarbon fields (like Barendrecht)
 - o Already thoroughly characterized and known past performance
 - o Existing wells allow access, but also imply integrity risks.

- Storage in operated oil fields, in combination with enhanced oil recovery
- Deep saline aquifer formations, preferably in the neighborhood of oil/gas reservoirs (like In Salah, Utsira, Schlepner etc)

CORA will not support the following storage types:

- Coalbed formations

2.2 Top risk events related to geological storage

2.2.1 Overview

The following top risk events can be identified, related to geological storage.

- 1 Unacceptable surface movement and induced seismicity at the surface, causing damage to people, installations and surroundings
- 2 Unacceptable leakage of harmful gasses at the surface level.
 - a This can be the injected fluid/gas (including generation by chemical reactions) Leakage failure criterion is for instance more than 1 % of injected gas leaks at the surface during 1000 years.
 - b Additionally: natural gas that has been released as a result of the injection.
- 3 Unacceptable contamination of groundwater or surface water by the injected fluid/gas or by migration of brine

2.2.2 Scope

CORA will support the geomechanical aspects of leakage.

2.3 Mechanisms and loading related to leakage failure

2.3.1 Leakage mechanisms

The following leakage mechanisms can be identified. These mechanisms are activated by different types of loading.

- Well bore failure during and after operation
 - o chemical interaction at the well bore (including corrosion and cement degradation)
 - o mechanical interaction at the well bore
 - o well head damage
 - o chemical interaction in a producing well
- Cap rock integrity failure:
 - o migration through the pore system in low-permeability cap rocks if the capillary entry pressure at which CO₂ may enter the cap rock is exceeded (over-pressurization)
 - o migration, because the cap rock is locally absent (natural pathways)
 - o lateral migration of free or dissolved CO₂ and incidental associated substances around seal into other fields (spilling);
 - o migration through faults (possibly reactivated) or other fractures in the cap rock.
 - o migration due to degradation of the cap rock or wells by reaction with acidic formation waters.
- Fractures, faults and natural pathways in the overburden structure.

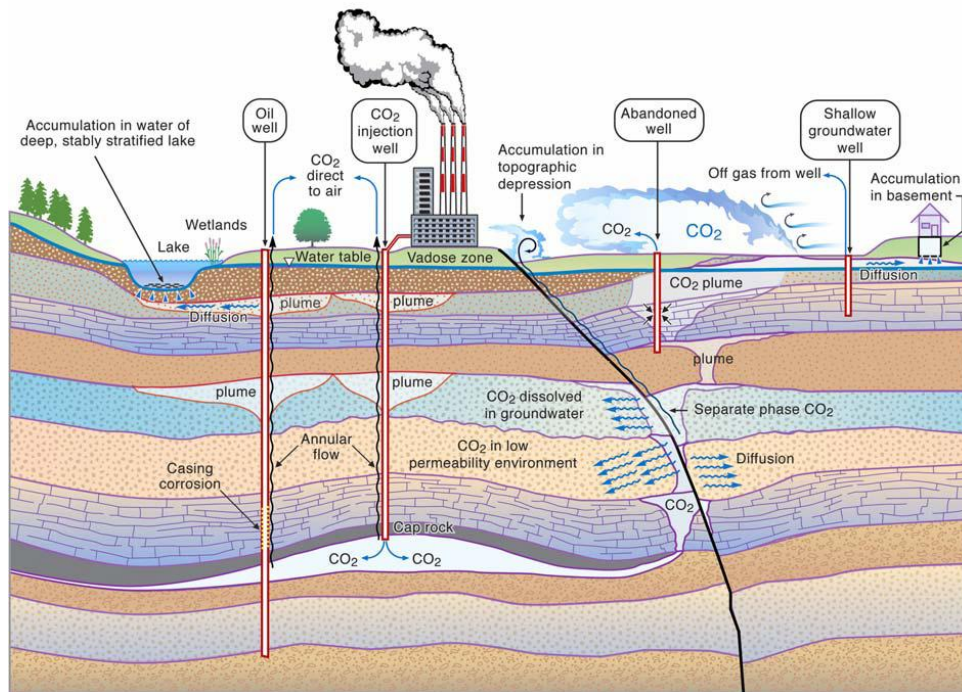


Figure 2.1: Schematisation of geological reservoir system for CCS [source: Oldenburg 2005]

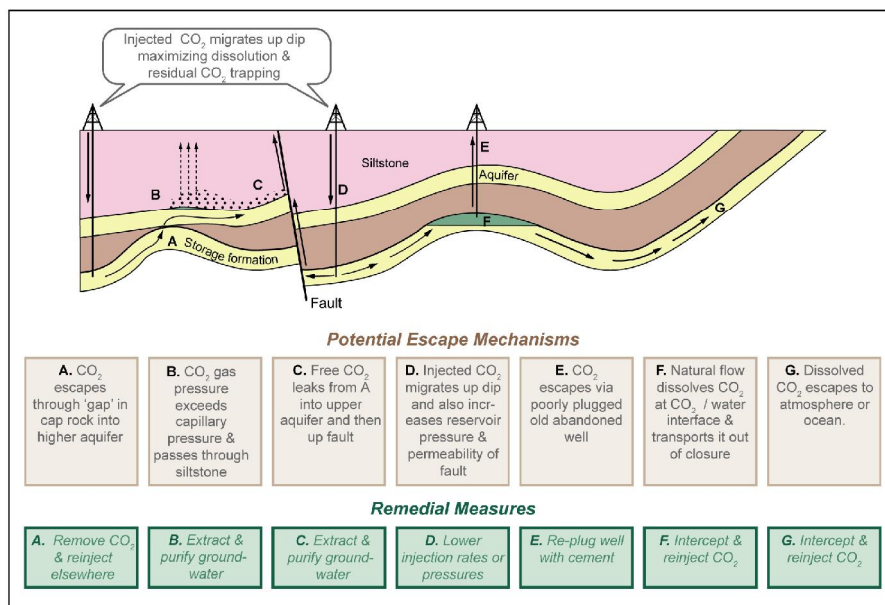


Figure 2.2: (IPCC, 2005)

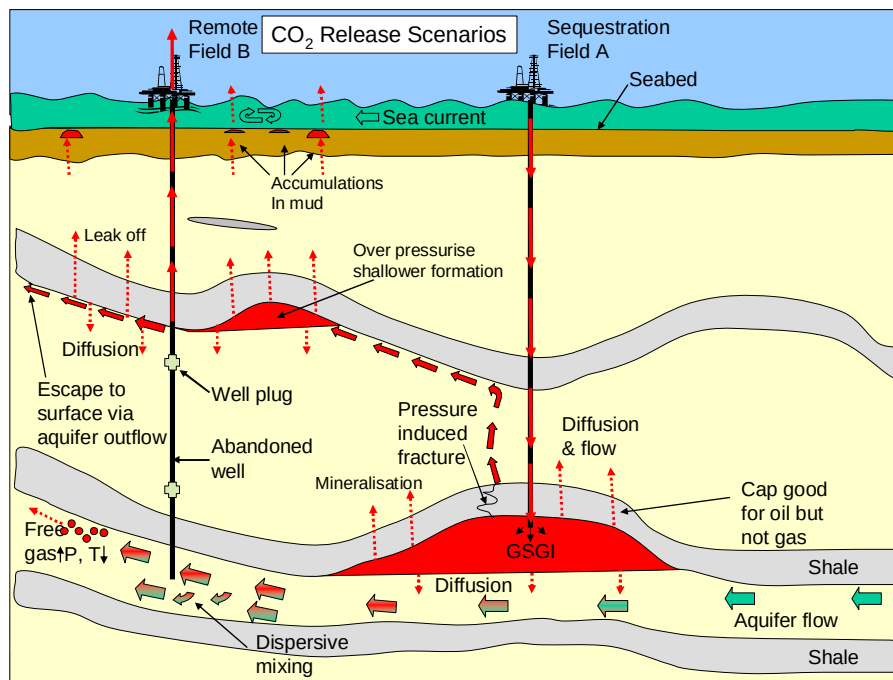


Figure 2.3: (CO2net)

2.3.2 Trapping mechanisms

The potential for leakage will mainly depend on well and cap rock (seal) integrity and other trapping mechanisms.

- *Cap Rock trapping*: Low permeable cap rock (primary seal)
- *Solubility trapping*: Dissolution in immobile capillary water/brine. A complete dissolution process of CO₂ is for example expected to take place on a time scale ranging from 10000 to 100000 yr.
- *Residual trapping*: surrounding capillary forces larger than gas/fluid pressure
- *Hydrodynamic trapping*: Long migration/leakage period of free or convected dissolved gas/fluid, due to thick overburden structure, with low permeability, no/few faults and additional seals. Fluid/gas injected in deep aquifers might take thousands to millions of years to migrate to the surface.
- *Chemical trapping*: mineral and ionic reactions and adsorption: will depend on the type of fluid/gas and the constitution of the reservoir.

Mineral and solubility trapping will increase in time.

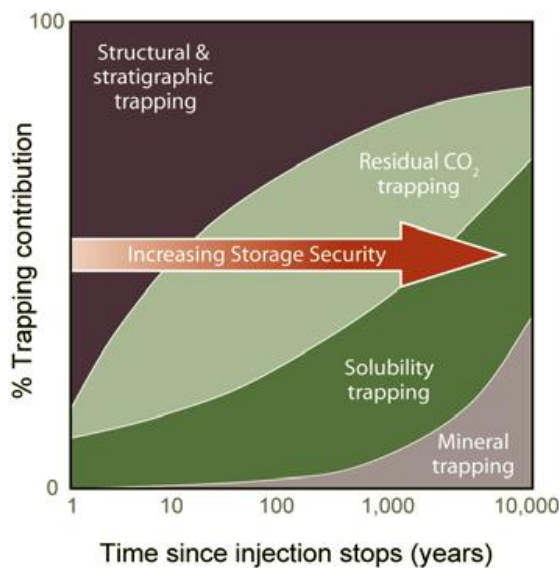


Figure 2.4: Trapping contributions versus time since injection

2.3.3 Scope

- CORA will consider the following geomechanical failure mechanisms.
 - o Fracture in caprock and overburden structure
 - o Fault failure in caprock and overburden structure
 - o Injection pressure exceeding capillary pressure
- CORA will consider the influence of injection pressures.
- The following loading types might become relevant in future.
 - o Geo-mechanical (stresses and strains)
 - induced micro-seismicity by injection
 - Tectonics and natural seismicity
 - induced micro-seismicity by drilling, mining etc.
 - o Thermal (temperature, fluxes).
 - o Hydrodynamic (non hydrostatic fluid pressures and discharges).
 - o Chemical (concentrations and reactions)
 - o Human intervention (drilling, mining etc). Causes also leakage when introducing pathways in the geological structure or when affecting active barriers (see next section on trapping).

- CORA will not consider well bore failure.
- CORA will not predict the amount of migrated gas/fluid, and therefore also will not consider trapping mechanisms

2.4 Supported Work Flow

2.4.1 Overview

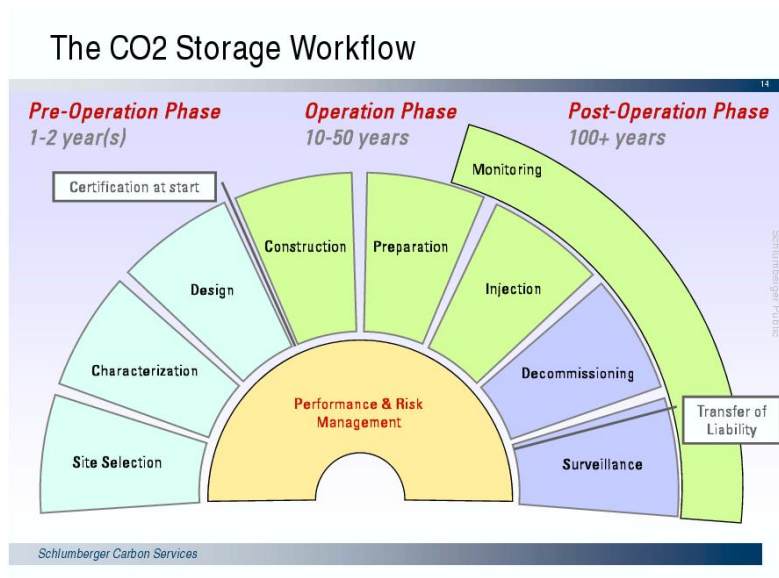


Figure 2.5: Overview of CO₂ Storage Workflow

2.4.2 Scope

CORA will support reliability analysis during the design, injection and storage stage, after a preceding detailed characterization and schematization. The degree of sophistication of the reliability model can be refined, depending on the stage and on the available data for characterization and schematization.

Excluded from the scope are site screening and selection based on rules of thumb.

2.5 Reliability analysis methods

2.5.1 Bayesian belief networks

Used to formalize expert knowledge and combine it with models. A nice feature of BBN's is that they can be used both forward and backward, e.g. to predict the probability of one or more events based on the probability of possible causes, as well as also to find the probability of causes in case of one or more occurring events (diagnosis). A network consists of nodes with one or more interdependencies. The status of each node in the network has conditional probabilities. This status can be for example true/false, but it can also be a value of a certain property, either in ranges or even continuous. The probability of the status of each internal or output node is calculated by using diagrams with conditional probabilities between the status of these node and the statuses of related other nodes. Such diagrams can be estimated by experts, but also by automated data mining. It is also possible to use a models for state prediction of a node.

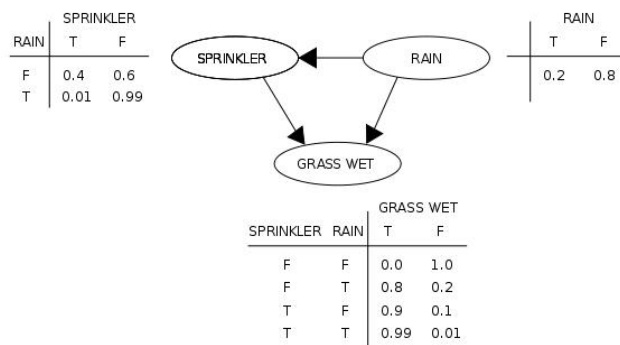


Figure 2.6: Example of a simple BBN

2.5.2 Probabilistic analysis

Probabilistic analysis can be used to calculate the probability failure at local level and at system level.

- Local failure occurs for a single mechanism at a single location.
- System failure occurs by failure for one or more mechanisms at any location.

The relationship between the probability of local failure for different mechanisms and different locations will be correlated, when they depend partially or fully on the same input parameters.

2.5.2.1 Local Limit State Functions

Local failure is defined by the condition that a certain solicitation value S becomes larger than an allowed resistance value R . The value $R - S$ is called the Limit State Function (LSF) or Z function.

$$LSF = R - S$$

For example: The fault shear capacity will be exceeded if an integrated shear stress along the fault length L exceeds the slip value (resistance).

$$LSF = \int^L \tau_{\max} - \int^L \tau$$

Limit State functions can be defined in different degrees of conservativeness. For instance: it also possible to define fault failure more conservatively, as the probability that the shear stress exceeds its allowed value, at any location along the fault.

$$LSF = \tau_{\max} - \tau$$

See section 3 for a specification the limit state functions to be used as part of CORA.

2.5.2.2 Proxy functions for the limit state

If solicitation values such as stresses, strains and displacements can be determined only with the help of a computational expensive prediction model, then so-called *proxy functions* for the LSF can be determined for computational efficient usage within a probabilistic procedure. Proxy functions are derived from a limited number of model predictions (scenarios), where each scenario represents a certain combination of parameter values.

In case of a linear dependency between the limit state function and different model parameters, linear proxy functions will resemble the true model response perfectly. The coefficient for each parameter variation is equal to the derivative of the solicitation value for variation of that parameter. The number of required scenario's to calculate the derivatives is therefore $n + 1$, where n is the number of parameters.

$$LSF = \sum J_i x_i = \mathbf{J} \mathbf{x}$$

Where

- x** the parameter vector, with parameters x_i
- J** the gradient vector of the linear LSF, where J_i is the derivative of the LSF with regard to parameter x_i .

In case of a nonlinear dependency between the solicitation value and the model parameters, regression techniques such as least squares could be used in theory to fit the coefficients of the selected regression function as close as possible on the model predictions of the solicitation values. Such a procedure is also known as “Experimental Design”. One should realize however that a response function which fits well on predicted values does not necessarily give an accurate prediction in the actual failure region. A smooth proxy function is only appropriate if the actual response in the failure region is equivalent smooth. A smooth response is however not to be expected in case of physical nonlinearity.

A typical set of scenarios for regression in case of a nonlinear response is generated by taking combinations of a lower and upper bound for each parameter, together with one combination of all mean values. The upper and lower values are usually related to the probability distribution of the parameter, by choosing a certain exceeding probability for the parameter value (for instance P10, P90). The number of scenario's to calculate regression coefficients is then $2^n + 1$. It is important to choose a proper nonlinear regression function. In practice, the function choice is often restricted to a regression model with second order terms (quadratic and/or cross terms). Analysis of variance should be used in any case to check the goodness of fit (residual error of the fit) and the significance of different regression coefficients.

Note: The results from an analysis with upper and lower bounds of all parameters can also be used for a semi-probabilistic safety assessment. This requires however the introduction of partial factors to assure that the allowed failure probability will not be exceeded.

Note: the model predictions for single parameter variations (with all other parameters at mean value) are commonly used for visualization in so-called Tornado diagrams, for the purpose of sensitivity analysis.

2.5.2.3 Limit State Surface

The *Limit State Surface* (LSS) is defined in the multidimensional parameter space by the combinations of parameters where the LSF is exactly zero. In case of a system analysis, a limit state surface will usually consist of contributions from different LSF's (different mechanisms at different locations).

2.5.2.4 Probabilistic input

Probabilistic analysis requires that probability distributions are known for the uncertain part of the model input. Continuous distributions are typically used for uncertain parameters and determined by a mean value, a distribution type and a standard deviation. Continuous spatial variability of physical parameters is sometimes included by introduction of a correlation model with a correlation length. Discrete distributions are sometimes used additionally for different geometry scenarios, where each geometry scenario has a probability of occurrence.

2.5.2.5 FOSM and FORM methods in case of a single LSF at a single location

FOSM (First Order Second Moment) is the name for an analytical method to derive the standard deviation of a solicitation value in a single LSF from the standard deviation of the parameters, assuming a normal distribution for all parameters.

If one assumes that the LSF depends linearly on each of the different parameters, then this analytical expression is as follows:

$$\text{LSF} = \sum J_i x_i = \mathbf{J} \mathbf{x}$$

$$\sigma^2(\text{LSF}) = \mathbf{J} \mathbf{C}_x \mathbf{J}^T$$

Where

- $\sigma^2(\text{LSF})$ the variance of the limit state function. The variance is the square of the standard deviation σ . This value is also the variance of the solicitation function, if the resistance is a deterministic value.
- $\mathbf{x}$ the parameter vector, with parameters x_i
- $\mathbf{J}$ the gradient vector of the linear LSF, where J_i is the derivative of the LSF with regard to parameter x_i
- $\mathbf{C}_x$ the covariance matrix of the parameters, where $\sigma^2(x_i) = C_{x,ii}$, and where the outer diagonal terms express the dependencies between the variations of different parameters (correlation coefficient times the standard deviations of the correlated parameters).

If the LSF is nonlinear, then the approximate *First Order Residual Method* (FORM) is often applied. FORM is an iterative FOSM method. FORM approximates the Limit State Surface by a linearization in the so-called design point. This design point is the combination of parameters on the Limit State Surface with the largest probability. FORM determines the design point iteratively, using a gradient search. The gradients are approximated numerically, using small parameter variations which are normalized with the standard deviation (u values). In u space, the design point is found as the limit state surface location with a minimum distance to the origin (mean).

A FOSM method (including FORM) is not suited for the simultaneous analysis of multiple mechanisms at multiple locations. Combination of results from different FOSM analyses for separate failure mechanisms at separate locations requires the use of approximate

combination methods (outcrossing or Hohenbichler), which become quite complicated in case of a dependency (correlation) between different LSF's at different locations.

An additional limitation of the FORM method is that the iteration procedure will not always converge from the given start point to the true design point in case of limit state surfaces with local minima in u-space.

2.5.2.6 Sampling methods in case of system analysis

The probabilistic methods Monte Carlo and Directional Sampling are suited for the integrated analysis of multiple failure mechanism at different locations.

The probability of failure is determined by these methods as the number of model predictions showing failure, divided by the total number of random model predictions. Therefore, these methods can supply not only a probability of failure, but also a full probability distribution of the different LSF's and corresponding solicitation values.

Monte Carlo uses a random sampling for all normalized parameters variations, whereas Directional Sampling uses a discrete number of fixed ratio's between the normalized parameter variations, in combination with random sampling of the scaled parameter variations. Directional Sampling is more efficient than crude Monte Carlo if the allowed probability of failure is low.

The required number of LSF determinations for a reliable result is a manifold of the inverse value of the allowed failure probability. Therefore, the required number will be very large if the allowed failure probability is very low. Efficient computation of the LSF is therefore a prerequisite for practical usage of sampling methods.

2.5.2.7 Adaptive Limit State Surface (response surface)

A response surface is a direct description of the Limit State Surface, allowing for efficient probabilistic analysis with sampling methods. Response surface equations include for example polynomials and splines, as well as indirect descriptions through neural networks.

In case of linear models, the response surface can be constructed a-priori, with the help of the exact linear proxy functions. This has been discussed in section 2.5.2.2.

In case of nonlinear limit state functions in combination with sampling, the use of a-priori determined response surfaces might not be sufficient, as the true model behavior at failure can differentiate from the response approximation. Using computational expensive model predictions during each sample evaluation is however not feasible in practice. In those cases, it is possible to set up and improve the response surface description as part of the sampling procedure itself.

The adaptive procedure starts with the construction of an initial response surface, by connecting a number of limit state points in the so-called u-space, where u is a parameter variation, normalized with the standard deviation. These limit state points are found by scaling the normalized parameter variations in a number of different directions.

The occurrence of failure for a certain combination of parameters in u-space is predicted during sampling by first using the response surface approximation. If the parameter combination sample is at close distance to the response surface, then an actual model prediction is performed. The result of the model prediction is finally used to improve the response surface description.

2.5.3 Scope

In combination with geomechanical models, CORA will support scenario analysis, sensitivity analysis and probabilistic analysis, the latter mostly with the use of intermediate proxy functions. CORA will not support other types of reliability analysis, including Bayesian Believe Networks.

The supported scope is detailed below.

- Automated scenario analysis, for the purpose of sensitivity analysis and for the purpose of setting up proxy-functions.
- Setting up linear and nonlinear proxy functions for limit state values and/or solicitation values (notably stresses, strains and displacements) at different locations, based on a limited number of model predictions, using a scenario table with different combinations of parameter and loading variations. Direct determination will be used in case of linear proxy functions. A least squares fit on regression functions with linear and quadratic terms will be used in case of nonlinear functions.
- Probabilistic analysis with FOSM/FORM for a single LSF at a single location. Probabilistic analysis with Monte Carlo and Directional Sampling for system behavior. Both capable of using proxy functions, as well as direct model predictions. Both based on the distribution input of the uncertain mechanical parameters and uncertain loading. Correlated spatial variation of physical properties will not be included. Uncertain geometry might be added by support of a number of geometry scenarios with associated probability. The analysis will be able to use proxy functions as well as direct model predictions. Output of FORM is the failure probability and the standard deviation of the solicitation function. Output of Monte Carlo and Directional sampling will be a full probability distribution of different solicitation values, as well as the probability of failure for different mechanisms at different locations.

The degree of sophistication of the geomechanical models can be refined, depending on the stage and on the available data for characterization and schematization.

2.6 Fault Tree Analysis and Event Tree analysis

2.6.1 Fault trees

A deductive fault tree relates certain top events (for instance loss of containment), through possible intermediate failure modes (such as leakage and lateral migration) to basic failure events (such as exceeding allowed stresses and displacements at different locations). A fault tree can therefore contain different levels, with AND gates and OR gates. A fault tree is generally used for schematization and visualization of the different contributions to a certain top event, including the dependencies (correlations) between these events.

Quantitative fault tree analysis can be used to calculate the probability of a top event, using calculated or estimated probabilities of the basic events. Quantitative fault tree analysis assumes mostly that the separately determined failure probabilities of basic events are independent (uncorrelated). Quantitative fault tree analysis is only required if there is no practical option for an integrated fully probabilistic analysis, for instance if an integrated probabilistic analysis would take too much computational effort, or if the fault tree contains conditional AND gates for different failure mode models.

The following conceptual fault tree (containing only OR gates) is presented hereafter for the following leakage mechanisms:

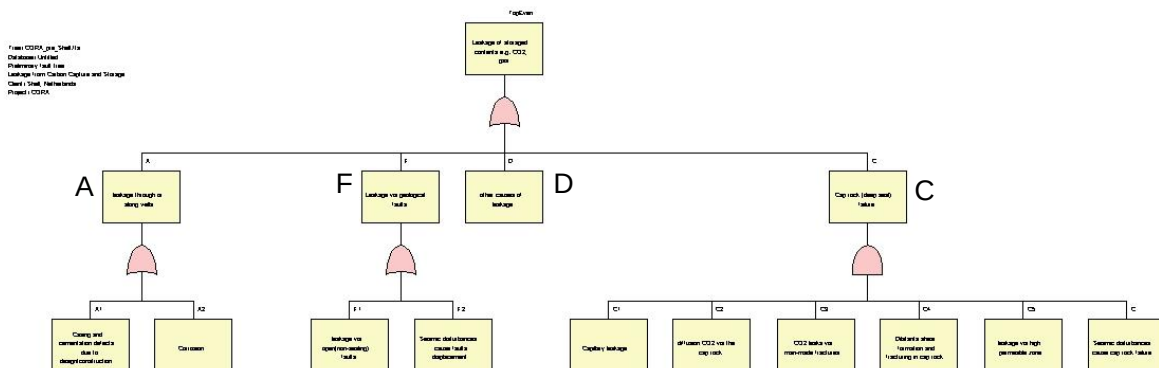


Figure 2.7: Conceptual fault tree of geological storage reservoir; top event: leakage of trapped content

A - CO2 leakage through or along wells

- A1 -Casing and cementation defects due to improper design/construction
- A2- Corrosion of casing and cementations

C- cap rock (seals) failure

- C1- Capillary leakage
- C2- Diffusion of CO2 through the cap rock
- C3- CO2 leaks through man-made fractures
- C4- Dilatants shear formation and fracturing in cap rocks
- C5- High-permeability zones
- C6- Seismic disturbances might cause cap rock failure

F- Fault: Geological failure

F1- CO₂ might leak through open (non-sealing) faults

F2- Seismic disturbances cause faults displacement (nature)

D- Other causes

The fault tree is preliminary constructed by putting these above failure mechanisms of leakage in logical ways. The current tree comprises of 3 levels:

Tree level 0: Top event: leakage of any stored contents e.g. CO₂, Gas, ...

Tree level 1: the leakage occur due to failure of any component of the reservoir system

Tree level 2 on: Failure of the components due to any potential failure mechanisms.

2.6.2 Event trees

An inductive event tree represents the subsequent evolution of consequence inducing events, typically taking top events from fault trees as input.

Two examples of event trees from literature are given below.

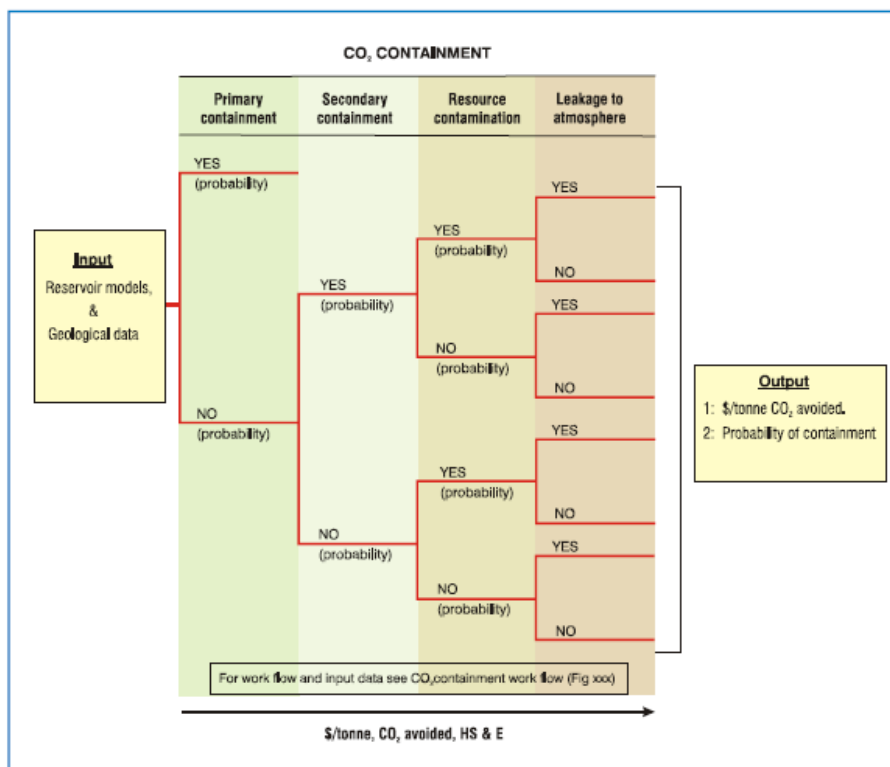


Figure 2.8: Example of event tree

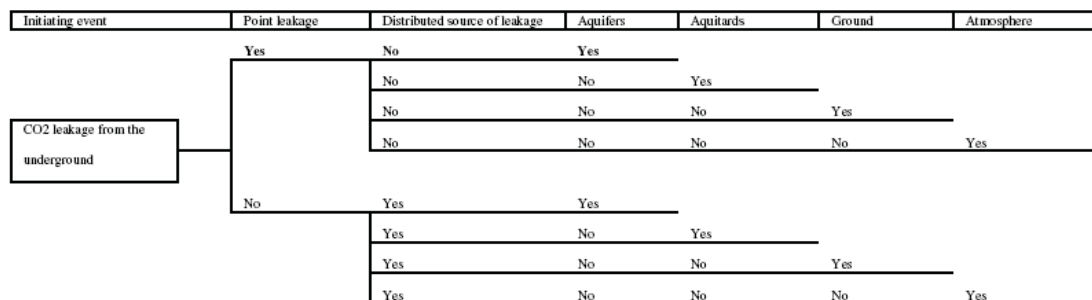


Figure 2: Example of consequence tree

Figure 2.9: example of event tree

2.6.3 Scope

The following assessment tree for CORA has been defined by Shell (vd Bogert, 2011):

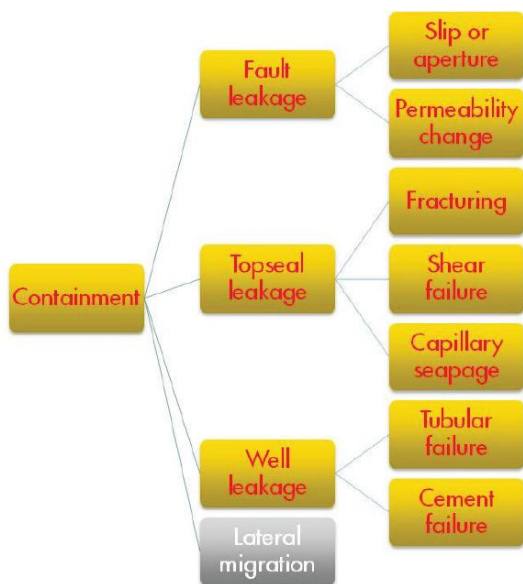


Figure 2.10: Assessment Tree according to Shell (vd Bogert, 2011)

Subsequently, geomechanical system failure has been defined as exceeding an allowable stress, strain or displacement at one or more locations. Chapter 3 gives an overview of associated limit state functions for fault, top seal and well failure.

When using a probabilistic sampling analysis, CORA can directly determine the system failure probability, by simultaneous evaluation of all limit state functions at all locations. The result from a CORA analysis can be used for example as input for a fault tree analysis, to combine with estimated or calculated failure probabilities for other failure modes.

2.7 Containment reliability analysis as part of risk management

2.7.1 Definition of risk

The generally accepted definition of risk is the occurrence probability of an adverse event, multiplied by the consequences resulting from this event.

Risk management follows in general the sequence of:

- Identifying risks;
- Quantifying the likelihood and severity of risks;
- Taking decisions on prevention and mitigation.

See also the picture below.

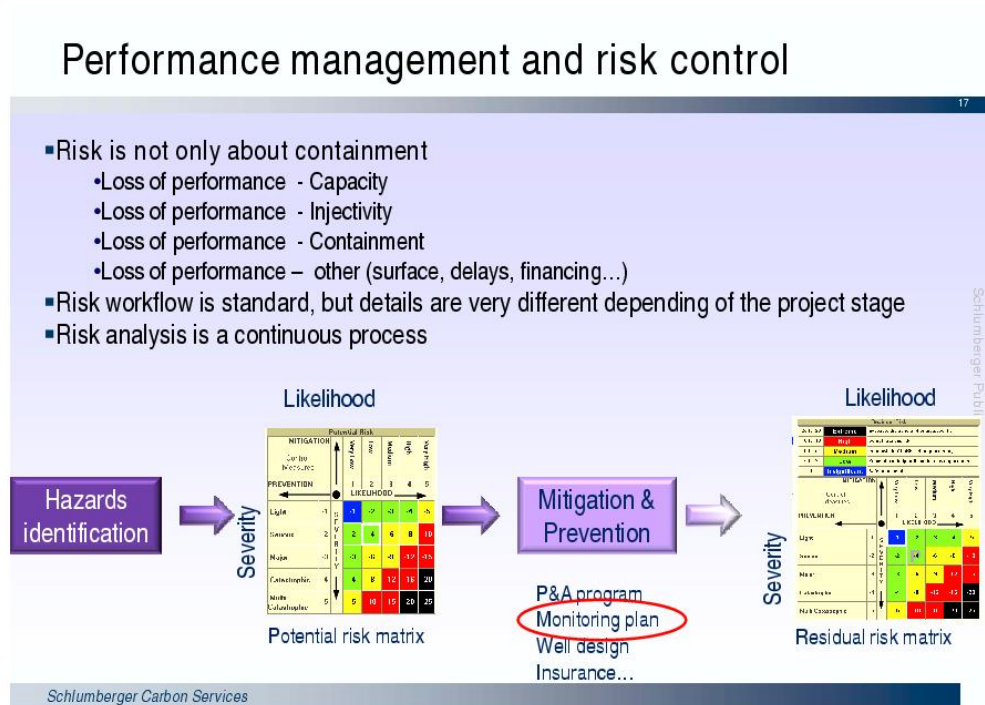


Figure 2.11: Risk management sequence

A full risk analysis must include all important risks (not just geomechanical) and, as stated before, take into account the failure probability as well as the failure consequences. The following figure illustrates that, after geomechanical failure, migration and reception should be considered additionally, before the consequences of leakage can be determined.

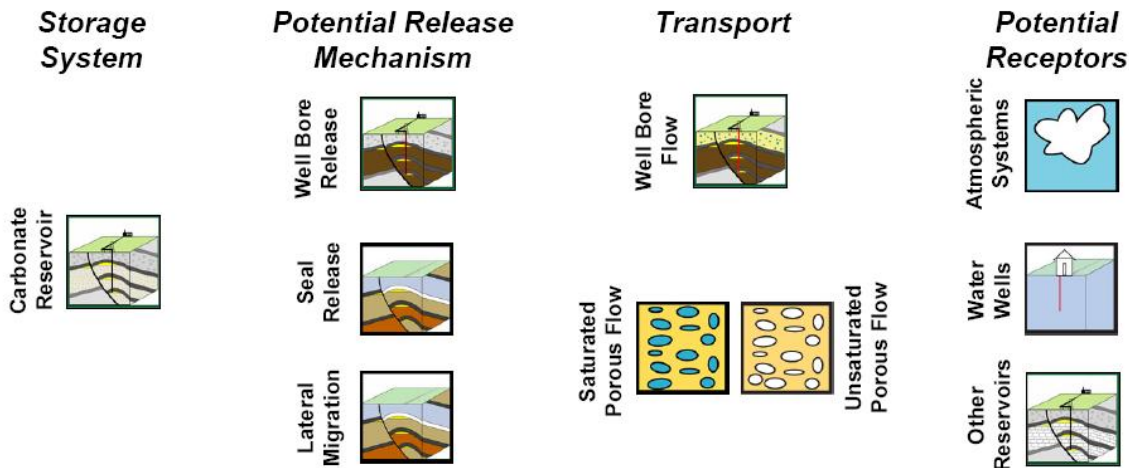


Figure 2.12: Source, Pathway, Receptor sequence for risk determination

2.7.2 Risk management methods

A recent overview of methods and literature for CCS risk management can be found here: http://www.cslforum.org/publications/documents/RATF_Phase1FinalReport.pdf.

Mentioned methodologies to support risk management are (a) Hazard and Operability Study; (b) Failure Modes Effects and Criticality Analysis; (c) Structured What-If Technique (d) Probabilistic Risk assessment and (e) Features, Events, Processes analysis. The latter can be used as a check-list for important potential risks and its causes. Quintessa has one [on-line](#). Relationships can be presented by different types of diagrams (Master Directed Diagram tree structure, Process Influence Diagram, Interaction Matrix).

2.7.3 Characteristics to consider for global risk analysis

The tables below give an overview of the characteristics to consider during global risk analysis of geological storage. Also check the [Quintessa FEP database](#).

High storage capacity	High porosity
High storage capacity	Large reservoir
Efficient injectivity	High permeability
Safe and secure storage	Low geoth. gradient & high pressure

Safe and secure storage	Adequate sealing over time
Safe and secure storage	Geological & hydrodynamic stability
Safe and secure storage	Minor adverse consequences of reservoir leakage
Safe and secure storage	Possibilities for monitoring, remediation and/or mitigation.
Low costs	Good accessibility, infrastructure
Low costs	Source close to storage reservoir

Table 2.1 (CO2Net + Ospar)

Characteristics	Attributes	Properties
Potential for primary containment	Primary seal	Thickness Lithology Demonstrated sealing Lateral continuity
	Depth	Distance below surface
	Reservoir	Lithology Permeability and porosity Thickness Fracture or primary porosity Pore fluid Pressure Tectonics Hydrology Deep wells Fault permeability
Potential for secondary containment	Secondary seal	Thickness Lithology Demonstrated sealing Lateral continuity Depth
	Shallower seals	Thickness Lithology Lateral continuity Evidence of seepage
Attenuation Potential	Groundwater hydrology	Regional flow Pressure Geochemistry Salinity

Table 2.2 (Oldenburg)

Table 8.1 Suitability of depleted gas fields for CO₂ storage

CRITERIA	Reservoir characteristics				
	Good		Medium		Poor
Existing and former penetrations					
Number of wells abandoned before 1985	0	0	1	2	3
Number of abandoned wells in total	0	2	4	7	>10
Number of wells to be abandoned	4	8	20	40	100+
Reservoir depth					
Depth of reservoir (m)	>2.500	1.000-2.500	800-1.000	500-800	<500
Faults					
Fault approach to surface (m)	>1.000	1.000 - 500	500 -150	150 - 10	< 10 (surface)
Number of faults	0	1-5	> 5	> 5	> 5
Assessed fault permeability	uniformly low		very uncertain		uniformly high
Cap rock					
Cap rock	salt	claystone (low permeability)	claystone (medium permeability)	other	other
Cap rock thickness (m)	> 100	50-100	30 - 50	< 30	< 30
CO ₂ in original gas phase in reservoir (%)	20	5	0	0	0
Reservoir rock					
Reservoir injectability	very good		medium		very tight
Reservoir rock	sandstone	carbonates	other	other	other
Earth tremor likelihood during injection (%)	0	10	50	100	100
Residual reservoir gas					
H ₂ S and BTEX in reservoir gas phase (%)	< 0.5	0.5 - 1.0	1 - 5	5 - 10	> 10
Surface features and uses					
Land use above reservoir	agricultural		nature reserve		urban
Proximity of vulnerable objects to potential leakage paths from the reservoir (m)	>100	50-100	25-50	10-25	<10

Table 2.3 (AMESCO)

2.7.4 Issues and parameters for detailed risk assessment

The following table from the OSPAR report gives an overview of the issues and related parameters to consider during detailed risk assessment of geological storage.

Issues to include in the risk assessment	Parameters to identify, qualify and –where possible- quantify in the risk assessment
Characterisation of the injected CO ₂ stream	Type and properties of other substances Concentrations of other substances
Location and geographical factors	Water depth, formation depth Human health and safety
Existence of amenities, biological features and legitimate uses of the maritime area	Areas of special ecological, economical or scientific importance, e.g.: – European marine sites – OSPAR MPAs – Sanctuaries – (Sensitive) species, communities or habitats – Breeding areas – Potable or irrigation water resources – Fishing areas
Regional geological setting	Regional geology, hydrogeology, hydrology, stratigraphy and structure Regional tectonics and seismicity Faults and fractures
Historical uses of the area	Man-made structures, including: – Integrity of active and abandoned wells with respect to CO₂ that are likely to be affected by the injection process - o Proximity to other wells (hydrocarbon producers, former or present) or fields - o Proximity to potable, irrigation or industrial water producing wells - o Proximity to other injection wells - o Age, depth and condition of the wells - o Geometry of plugs and casing and composition of plugs of abandoned wells Conversion of existing well for injection: information is needed on well age, its construction details, and its history

Reservoir/seal evaluation	Geological interpretation – Stratigraphic interpretations and well-log cross sections of the reservoir intervals – Reservoir/seal heterogeneity – Temperature, pressure, fluid characteristics (salinity) Geophysical mapping – 3-D maps of potential migration pathways (faults) – Structure and thickness of formations and cap rocks Petrophysics – Permeability, relative permeability (injectivity) – Porosity – Capillary pressure – Mineralogy Hydrodynamics – Displacement of formation water – Vertical hydraulic gradient Sealing capacity of cap rocks – Seal thickness – Capillary entry pressure Faults – Location, orientation and properties of faults or fractures that are likely to intersect the formation Geomechanics and geochemistry – CO₂ stream – water – rock interaction – Stress, stiffness and strength – Potential of the injected fluid to cause plugging of the formation – Compatibility with injected formation chemistry – In-situ stress profile in the various layers Other components in the input-stream Reservoir simulations – Short-term behaviour: formation response (pressure changes for a given injection rate) – Long-term behaviour: formation containment – Sufficient capacity of the formation for planned CO₂ storage Data quality – History, current status and age of information available on the geological formation
---------------------------	--

Marine environment characterisation	Ocean current and sea floor topography in the region Physical, chemical and biological characteristics of seabed, sediments and overlying waters: – Natural fluxes of CO₂ in the seabed and across the seabed surface – Chemical characteristics of the seawater – Nutrients and other substances (potential contaminants/pollutants) – Biological communities and biological resources o composition, structure, dynamic
Economic/regulatory factors	Economic feasibility Impact on other sub-seabed resources such as oil and gas extraction and other natural gas/CO₂ storage sites Regulatory framework Applicable regulations, codes and standards, and regulatory restrictions and restraints

Table 2.4 (OSPAR)

2.7.5 Tesla

Shell uses [Tesla](#) as a risk management and decision support tool, during different stages, starting from site screening and selection. Tesla uses a tree structure and evidence supported reasoning to capture expert knowledge.

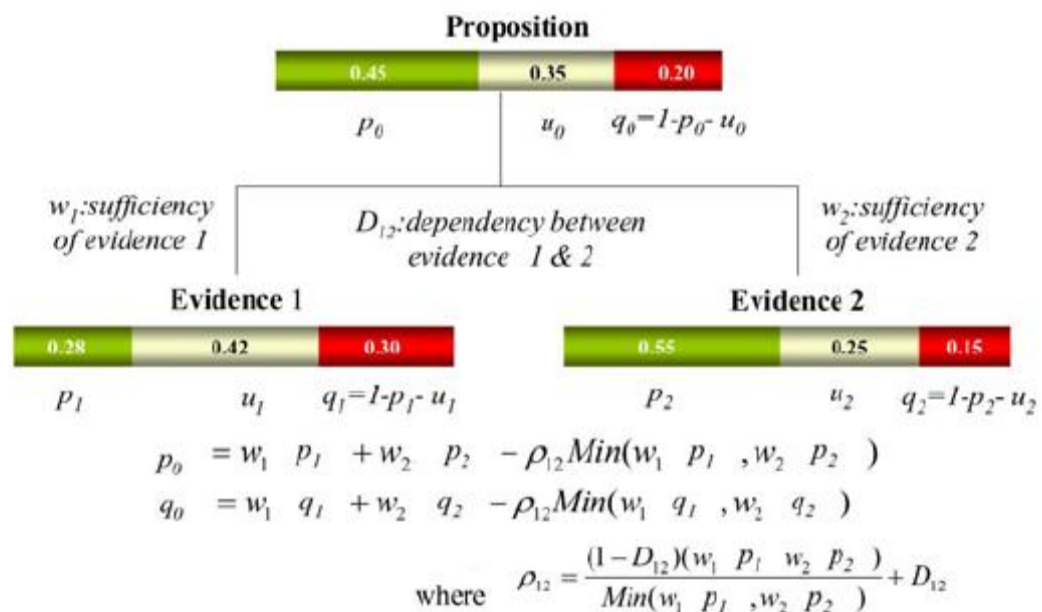


Figure 2.13: Illustration of Tesla's evidence based logic

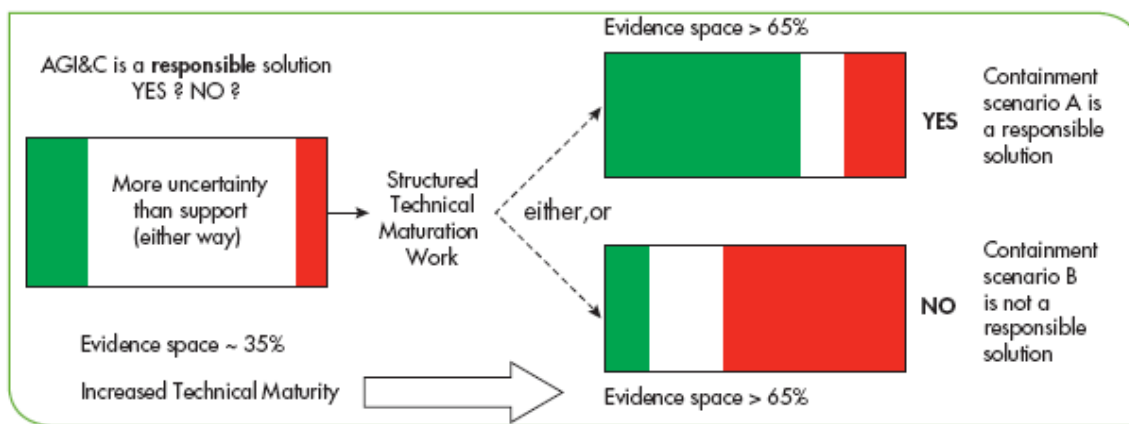
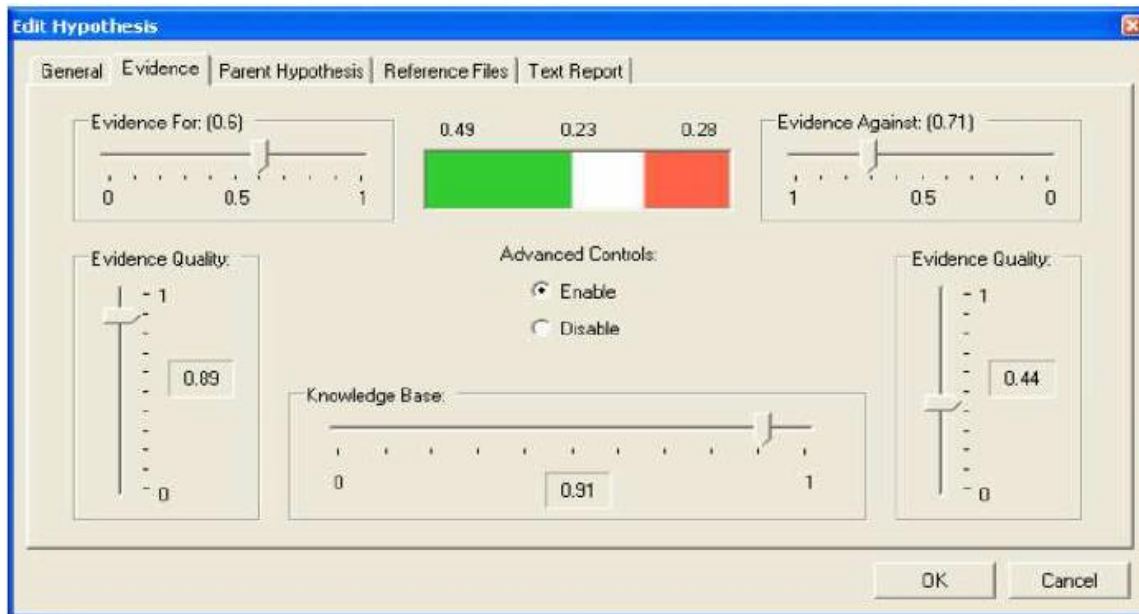


Figure 2.14: Example of using Tesla for guiding uncertainty reduction and for decision support

Main tree structure

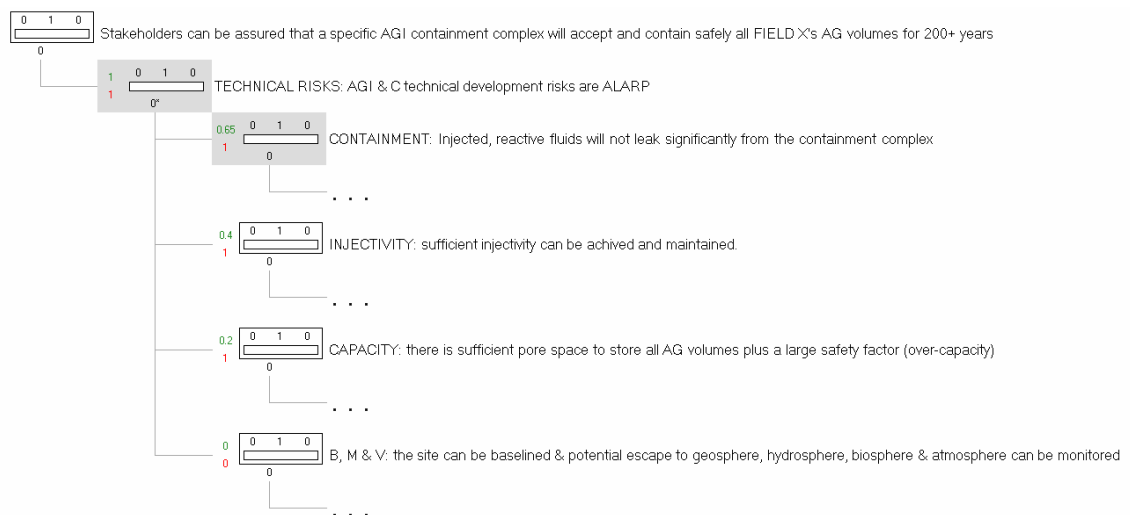
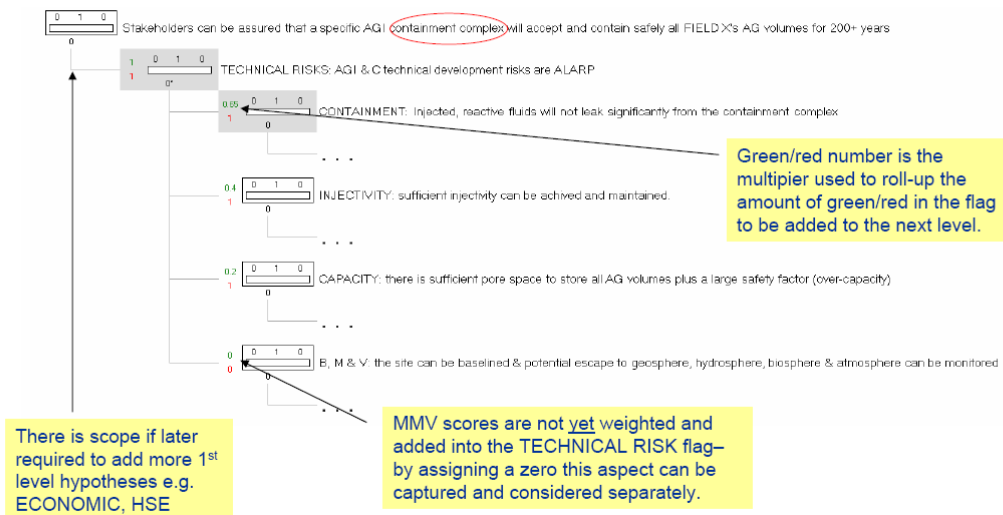


Figure 2.15: Example of Tesla tree structure

2.7.6 Scope

The aim of a CORA analysis is to contribute to Shell's risk management strategies during different project stages. CORA will support the execution of sensitivity and scenario analysis, as well as a probabilistic analysis. Probabilistic analysis yields the probability of a safe and an unsafe state, optionally with an intermediate uncertain state, using different safety criteria for different mechanisms and models. For a full risk analysis, the probability of geomechanical failure should be coupled in principal with probabilistic analysis of well integrity, reactive flow, gas migration etc through estimations and/or calculations with other models.

2.8 Relationship between reliability analysis and monitoring

2.8.1 Objective of monitoring

A summary of monitoring objectives (from the Amesco report) is presented below.

The overall objective of monitoring is to gain increasing evidence that the CO₂ storage is not leaking. Monitoring also helps to identify priorities in data acquisition and, if a leak has been identified, to take timely remedial action. Within the framework of a storage project, monitoring can be performed for many other reasons, ranging from meeting regulatory obligations and certifying CO₂ storage for emission trading schemes to informing the public and testing of new monitoring technologies. The intensity of monitoring for safety can be increased or reduced depending on the fit between modelled and actual behaviour of the storage site throughout the life of the project from the pre-operational phase through to the post- closure phase.

If the objective of monitoring is to gain increasing evidence that the CO₂ storage is not leaking, then the role of modeling as part of a learning and monitoring program can be considered. As part of the monitoring program, quantitative models should be constructed, but should not be seen as tools to make absolute statements about the predictions made. Rather, the added value of modeling is in learning during the injection and post-injection phases. This should allow the operator to steer his activities so as to optimise his operations within certain constraints (e.g. improving the risk profile in time as new information is acquired). Both optimization objectives and constraints are subject to change as the new truth is continuously being revealed in time (i.e. through the monitoring activities). Monitoring programs also need to be updated and re-focused as the new information is being revealed. The quantitative models are also useful to establish the priorities of a data acquisition (monitoring) strategy by 'Value of Information' (VoI) analyses. A VoI analysis predicts the impact of the storage operations with and without the information gained, assuming that the value of any new information entity is determined by how it impacts on subsequent decision-making (e.g. timing of remedial action). According to the VoI approach, if new information has no impact on the decision-making, then the VoI is zero.

See the [Amesco report](#), Chapter 9 for a detailed description.

Objectives of the monitoring plan - Summary

34		
Operational Monitoring	Verification Monitoring	Assurance Monitoring
Injection operation control - Wellhead pressure - Bottom hole Pressure and Temperature - Injection rate - Microseismicity	Well Integrity - Annulus pressure - Corrosion - Cement - Soil gas measurements	Impact: HSE monitoring - Potable water quality - Soils acidity - Atmospheric concentration - Surface deformation
	Cap Rock / Fault Integrity - Microseismicity - Pressure interference	Detection of leaks/migration - Sampling & chemical analysis - Geophysics techniques - Pressure interference - Soil gas measurements - Vegetation stress - Eddy correlation tower
Quantification of injected CO₂ - Mass flow - Gas stream composition and phase	Tracking the CO₂ plume - Geophysics techniques - Pressure, Temperature - Well logs (CO₂ Saturation) - Sampling - Geodetic methods	Quantification of leaks - Soil gas measurements - Surface gas measurements ...
Schlumberger Carbon Services		

Figure 2.16: Objectives of monitoring

2.8.2 Monitoring techniques

The following tables give a global overview of monitoring techniques for different targets. For a monitoring selection tool, see

http://www.co2captureandstorage.info/co2tool_v2.1beta/co2tool_panel.php.

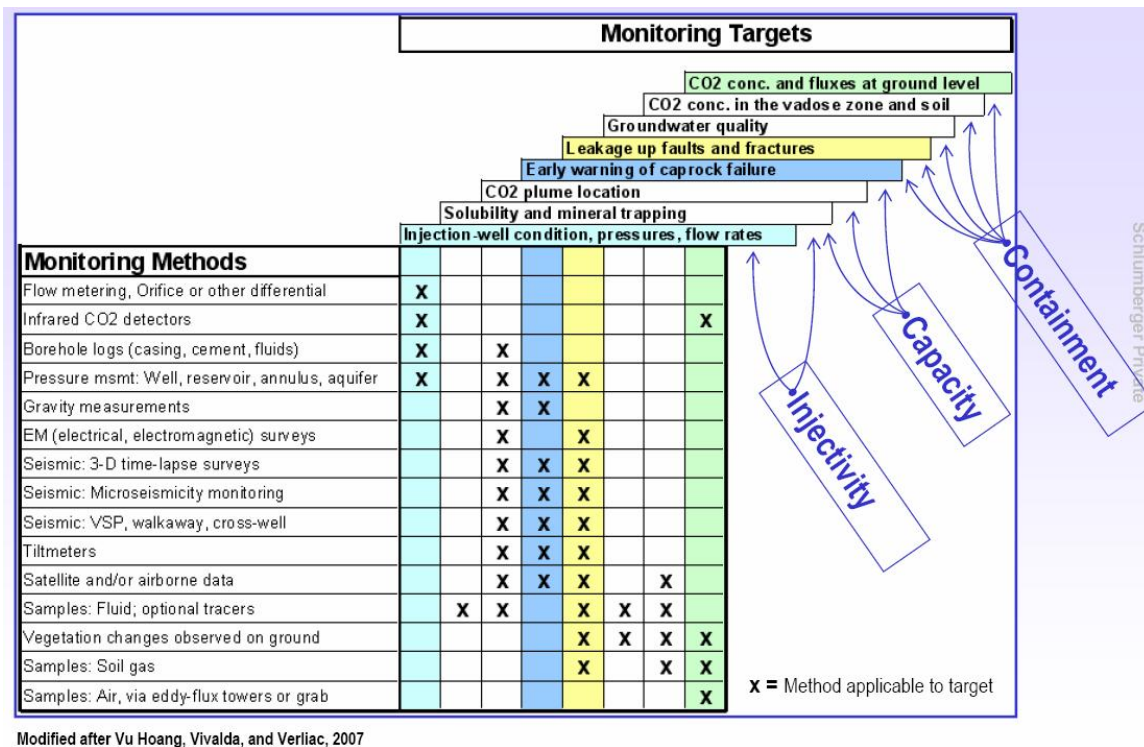


Figure 2.17: Monitoring techniques versus monitoring targets

Leakage path	Monitoring techniques	Application in monitoring phase
Pathways between cement, casing and formation	Logging tools (CBL, CIL etc.)	Preparation, operation, closure
	Soil gas sampling at the surface near the well.	All
Casing failure and through well	Logging tools, well fluid samples	Preparation, operation, closure
Pathways along well, into overburden	(Time-lapse) seismic surveys	Preparation, operation, post-closure
	VSP, reverse VSP	Preparation, operation, post-closure
Pathways along well, into overburden and to surface	Groundwater fluid sampling; soil gas sampling Resistivity methods at surface	All

2.8.3 Scope

The results from a probabilistic analysis support straightforwardly the selection of the required monitoring type, as well as the optimization of the locations and number of monitoring instruments, by presenting the locations with a high probability of local failure and by showing which parameter uncertainties yield the highest contribution to the local and total probability of failure.

Furthermore, CORA will obviously be able to use the measured injection pressures to give improved reliability estimates during the operational stage.

Usage of monitoring data for status-update of the stress and strain field and for parameter-estimation is not in the scope of a first version, but can certainly be added

later on. Especially in case of geological storage in depleted gas fields, when there is already an abundance of monitoring data available in the design stage. This can be achieved through the use of different data-assimilation techniques, such as weighted least squares in combination with Bayesian Updating or with Extended Kalman filters. The open source data-assimilation toolbox [OpenDA](#), developed by Deltares and others, contains such techniques. Data-assimilation implies that the results of stress, strain and displacement monitoring are be used to update the stress and strain field, as well as to improve the initial assumptions on values and distributions of (correlated) mechanical parameters.

3 Limit state functions for different failure modes

3.1 Introduction

As stated in chapter 2, the scope of CORA will be limited to the analysis of geomechanical failure modes. This means failure modes determined on the basis of stress, strain or displacement. The following failure modes and limit state functions are envisaged.

3.2 Fault shear failure

One or more of the following limit state functions should be evaluated for all faults in geometry that can cause leaking. For limit state functions evaluated in points, a practical simplification is to use only the most critical location following from a mean value analysis.

3.2.1 Stress criterion, based on failure initiation in one location

$$LSF = f_{allowed} - \frac{\tau}{\tau_{max.MC}}$$

Where

τ	the shear stress in any position along a fault
$\tau_{max.MC}$	the maximum shear stress in the same position, according to Mohr Coulomb. This value is derived from the normal stress and the Mohr Coulomb parameters in the fault (cohesion and friction angle)
$f_{allowed}$	The allowed utilization ratio of the shear stress

3.2.2 Stress criterion, based on final shear capacity along the fault

$$LSF = f_{allowed} - \frac{\int^A \tau}{\int^A \tau_{max.MC}}$$

Where

A	The area (3D) or length (2D) of a fault
$f_{allowed}$	The allowed utilization ratio of the shear stress capacity

3.2.3 Stress criterion, based on slip area

$$LSF = f_{allowed} - \frac{A_{slip}}{A}$$

Where

A_{slip}	The area where slip failure occurs
$f_{allowed}$	The allowed ratio of the fault area that is slipping

In linear elastic analysis the slip area will be underestimated if one assumes that this area is equal to the area where the stress criterion has been exceeded.

A better estimate for the slip length l_{slip} in case of 2D linear elastic analysis is achieved by searching the slip length where the following condition is fulfilled: $\int^{l_{slip}} \tau = \int^{l_{slip}} \tau_{max.MC}$.

3.2.4 Slip displacement criterion (nonlinear)

$$LSF = u_{allowed} - u_{slip}$$

Where

u_{slip}	The occurring relative slip displacement along the fault
$u_{allowed}$	The allowed relative slip displacement

Optionally, the allowed slip displacement can be related to an allowed relative dilatancy displacement, by using the dilatancy angle.

3.3 Top seal fracturing

Top seal fracturing should theoretically be checked in the total top seal volume. The failure criteria are based on principal stress values. As a practical simplification, the check could be limited to one or more critical locations (fracture hot spots), following from a deterministic mean value analysis.

3.3.1 Shear stress criterion in a critical location, based on failure initiation

$$LSF = f_{allowed} - \frac{\tau}{\tau_{max.MC}}$$

Where

τ	the shear stress in principal axes
$\tau_{max.MC}$	the allowed maximum shear stress in the same position, according to Mohr Coulomb. This value is derived from the stress tensor and the Mohr Coulomb parameters in the fault (cohesion and friction angle)
$f_{allowed}$	The allowed utilization ratio of the shear stress

3.3.2 Principle stress criterion in a critical location, based on failure initiation

$$LSF = \sigma_i - \sigma_{allowed}, \text{ or (if } \sigma_{allowed} \neq 0): LSF = f_{allowed} - \frac{\sigma_i}{\sigma_{allowed}}$$

3.4 Seepage through top seal

Top seal seepage occurs if the injection pressure exceeds the capillary pressure.

$$LSF = f_{allowed} - \frac{P_{injection}}{P_{capillary}}$$

3.5 Straining and shearing along a well in the top seal

3.5.1 Shear stress criterion

$$LSF = f_{allowed} - \frac{\tau}{\tau_{max.MC}}$$

3.5.2 Axial strain criterion

$$LSF = |\varepsilon_{allowed}| - |\varepsilon|$$

4 Limit state calculation

4.1 Proxy functions for LSF's, in case of linear elasticity

A probabilistic analysis requires numerous evaluations of different LSF values at different locations. In section 2.5.2.2 has been described already that it possible to use proxy functions to reduce the computational costs of LSF evaluation. Using proxy functions is especially advantageous if the stress model is linear elastic, because then the proxy function for stress, strain and displacement will be linear also.

4.1.1 Linear response of stress components

In case of a linear elastic stress model, each component of an incremental displacement vector, strain tensor and stress tensor will vary linearly with increments of material parameters and loading. Therefore, it is possible to write the components of a stress or strain tensor as a linear function of the parameters x_i , ($i=1,n$) with matrix **J** as a coefficient matrix.

$$\begin{pmatrix} \Delta u_x \\ \Delta u_y \\ \Delta u_z \end{pmatrix} = \mathbf{J}_u \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \Delta x_4 \\ \Delta x_5 \\ \cdot \\ \cdot \\ \Delta x_n \end{pmatrix}, \quad \begin{pmatrix} \Delta \varepsilon_{xx} \\ \Delta \varepsilon_{yy} \\ \Delta \varepsilon_{zz} \\ \Delta \varepsilon_{xy} \\ \Delta \varepsilon_{yz} \\ \Delta \varepsilon_{zx} \end{pmatrix} = \mathbf{J}_\varepsilon \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \Delta x_4 \\ \Delta x_5 \\ \cdot \\ \cdot \\ \Delta x_n \end{pmatrix}, \quad \begin{pmatrix} \Delta \sigma_{xx} \\ \Delta \sigma_{yy} \\ \Delta \sigma_{zz} \\ \Delta \sigma_{xy} \\ \Delta \sigma_{yz} \\ \Delta \sigma_{zx} \end{pmatrix} = \mathbf{J}_\sigma \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \Delta x_4 \\ \Delta x_5 \\ \cdot \\ \cdot \\ \Delta x_n \end{pmatrix}$$

This linearity is preserved in any direction of vector or tensor transformation.

4.1.2 Linear response of LSF's for faults and wells (fixed direction)

The LSF value will therefore also be a linear function of parameters, as long as it only uses linear combinations of components of stress, strain or displacement, with each component expressed in fixed directions. This is true for all currently identified failure criteria in faults and wells. Exact linear functions for these LSF's can therefore be derived straightforwardly.

4.1.3 Nonlinear response of LSF's for top seal failure (arbitrary direction)

The two LSF's for top seal fracturing are expressed in principal stresses. These principal stresses with variable direction are a nonlinear function of the different stress components in fixed directions.

Instead of trying to derive direct approximate proxy functions for these LSF's, it seems advantageous to use exact linear proxy functions for the different stress components, and execute the only remaining calculation of the nonlinear principal stress from these stress components during a probabilistic procedure suited for nonlinear LSF's.

4.1.4 Strategy for accurate and efficient LSF evaluation with proxy functions

Summarizing, an efficient limit state function evaluation as part of probabilistic procedure has to follow the follow logic:

- If the LSF is based on stresses/strains/displacements in a fixed direction and if the stress model is linear elastic: calculate the LSF value with the exact proxy function.
- If the LSF is based on principal stresses and if the stress model is linear elastic: calculate the stress components in the LSF with the exact proxy functions for each of these stress components.
- If the LSF is based on a nonlinear stress model: use approximate proxy functions for either the full LSF or for different stress components in the LSF, as long as the predictive quality of these proxy function(s) is sufficient.

4.2 Interface between CORA and external limit state models

It is envisaged that CORA will connect to external software for the calculation of the limit state function values. The interface between CORA and this software will contain the following input and output.

4.2.1 Stress model input or proxy function input

- Calculation with stress models. The limit state models require full input for the attached stress model. For the static input of external stress models (geometry, boundary conditions, initial stress field etc) however, it is foreseen that CORA will only point to a separately supplied input file. The API of the interface with CORA will be limited to the additional variable part of the stress model input.
- Calculation with Proxy Functions. In case the limit state functions have to be evaluated with proxy functions for LSF's or stress components, then the proxy table is input.

4.2.2 Additional input for calculation with stress model

- Per fault:
 - Selected LSF type
 - The resistance value corresponding to the selected LSF type (allowed shear utilization ratio, allowed slip area utilization ratio or allowed displacement)
 - Location (2D: center, dip angle, length)
 - Strength properties (cohesion, friction angle)
- Top seal (stress components only):
 - The user-defined "hot spot" locations for calculation of the stress components
- Per Well
 - The resistance value corresponding to the LSF type (allowed strain).
 - Location

4.2.3 Additional input for calculation with stress proxy functions for top seal

- Selected LSF type

- The resistance value corresponding to the LSF types (allowed shear utilization ratio, allowed capillary pressure utilization ratio)
- Strength properties (cohesion, friction angle)

4.2.4 Output

Output of the limit state software is a full list of LSF/stress component values for all selected LSF types and all defined locations.

4.3 Limit state calculation with stress models

Three different models are in view to predict the stresses, strains and displacement in the LSF's defined in chapter 3. These models are, in order of complexity: Geertsma, QuickBlocks and Geomec/DIANA.

	Geertsma	QuickBlocks	Geomec/DIANA	
Geometry	Axy-symmetric. Single disc	3D, Multiple blocks	3D, Arbitrary	
In-situ stress	Homogenous, constant gradients	Layered with constant gradients	Realistic	
Formation properties	Homogenous, linear-elastic	Layered, linear- elastic	Arbitrary, linear- elastic	Arbitrary, non- linear
Reservoir scenario	Single delta P	Delta per block	Realistic	
Results location	Arbitrary point	???	Calculation nodes	

Table 4.1 Stress models

Input for the Geertsma model is:

- Disc geometry (depth, radius, height)
- In-situ stress field (vertical gradient, horizontal gradient)
- In-situ pore pressure field (vertical gradient, horizontal gradient)
- Rock properties(density, young's modulus, poisons ratio)
- Reservoir depletion pressure
- Coordinate of the point to calculate stresses, strains and displacements

The input for Quickblocks and Geomec/Diana is more complicated, especially with regard to the definition of geometry, boundary conditions and loading. This static input will be contained in a separate input file.

The limit state software requires from the stress modes as a result the values of stress, strain or displacement components for a number of specific locations.

- In case of Geertsma, this output can be derived directly, using the analytical equation. The limit state software will contain the model itself and therefore directly supply the required output of stresses, strains and displacements in all required locations.

- In case of GeoMec and Quickblocks, each parameter scenario requires the execution of a single numerical field prediction, followed by a mapping of the numerical results to the required output locations. The limit state software will only act as a layer that prepares the stress model input, call the stress model once per parameter scenario, and if required, performs a mapping of the numerical result field to all locations required by CORA. The input and output of Quickblocks and Geomec for each of the analyzed parameter scenarios can be saved on disk and can therefore be inspected and used also independent from CORA.

5 Definition of CORA, version 1

5.1 Introduction

Section 5.2 cites the vision of Shell on three different stages of maturity in the development of containment reliability assessment framework. The sections thereafter contain a proposal for the definition and design of CORA version 1, which is identified hereafter as a *lab tool*, with as its main purpose to supply a proof of concept.

5.2 Development stages

This section cites the vision of Shell (according to vd Bogert, 2010) on three different stages of maturity in the development of containment reliability assessment framework. The associated PC tools are identified as *lab tool*, *stand-alone tool* and *integrated tool*. The user interface and probabilistic calculation engine will have different attention in the three maturity stages as elaborated below.

- 1 The lab tool is for specialists to provide proof of concept and to test the approach on realistic case studies. This version of the tool focuses on providing functionalities for automated sensitivity analysis and probabilistic analysis. Therefore, key objective of this stage is to develop a re-usable calculation engine and test all relevant approaches. At least two major releases of the lab tool are foreseen: one with basic and a second release with more advanced functionalities. This second release will include approaches to constrain the probabilistic evaluation with field observations and to extend the deployment scope of the tool by improving computational efficiency. Experimental Design – a technique to reduce computational efforts – is already being tested and will therefore already be included in the first release. Current expectation is that the engine needs re-development when integrated with reservoir engineering approaches.

The user interface of the lab tool will assume “Excel-level” skills of the users (Matlab and Python would require dedicated training and experience). The deterministic calculation engines (Geertsma and QuickBlocks) determine input requirements, whereas output is mostly numerical with extended graphical support brokered through Microsoft Excel. The user interface will be abandoned in the stand-alone stage.

- 2 The development of the stand-alone tool will focus on a user interface for deployment by (trained) project engineers in an independent fashion. Requirements for input and output, analysis options, speed, and interfaces with deterministic engines should be obtained from the lab tool stage. Interfaces with (operations) databases and other analysis result will be matured in this stage. A web-based user interface with server-client architecture is foreseen as it facilitates parallel processing of the computations.
- 3 The development of an integrated tool will focus on incorporating more sophisticated deterministic calculation engines. This could comprise of some form of coupled geomechanical & reservoir simulation (reactive flow simulation). Operational failure criteria would probably need reassessment, whereas a uniform probabilistic approach for reservoir and geomechanical assessment needs to be implemented. Re-development of the reliability assessment tool as part of a single system (DRMS/SRMS) would likely be required.

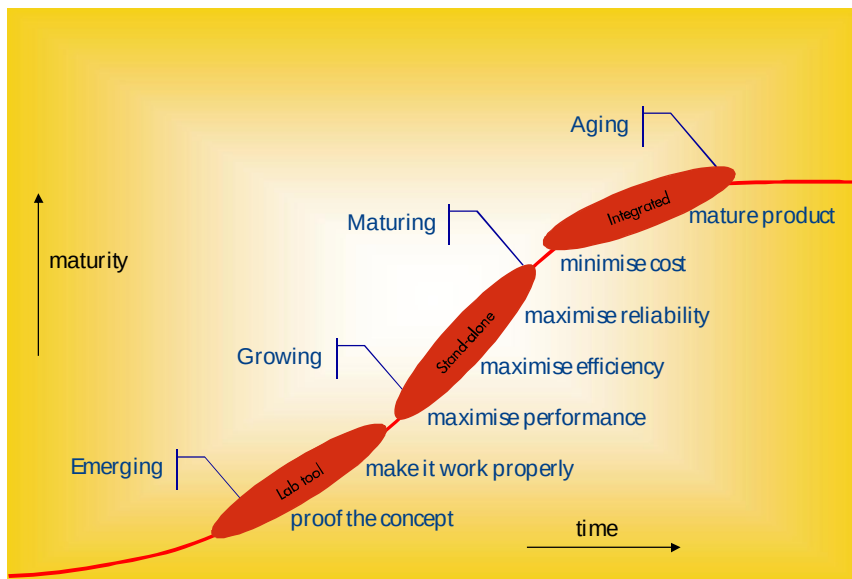


Figure 5.1: CORA development stages, according to Shell

5.3 Summary of user requirements for CORA version 1

- 1 Applicable for geomechanical sensitivity and reliability analysis of reservoir injection and depletion.
- 2 Limit state assessment in fault, top seal and well locations, using failure criteria based on stress, strain and displacement.
- 3 Connection to three stress models (Geertsma, Quickblocks, Geomec).
- 4 Support for scenario analysis, with the following two purposes:
 - a preprocessing for the automated determination of limit state proxy functions
 - b sensitivity analysis, with output of Tornado diagrams, as well as graphs of different Limit state function values versus different parameter values
- 5 Automated determination of proxy-functions for limit state values and stress component values
- 6 Efficient linear elastic probabilistic analysis, by evaluation of limit states through the use of linear proxy-functions
- 7 The ability to define/add/alter/configure limit state models and attached stress models, independent of the CORA framework
- 8 Only input through CORA of the variable part of external stress models. The static part will be contained in externally supplied input files.
- 9 Usage in a MS- Windows environment
- 10 Rudimentary graphical user interface, with the ability to export results to Excel.

5.4 Definition of functionality for CORA version 1

Hereafter follows a schematic overview of functionality for version 1. The part that is not required for linear elastic stress models (nonlinear proxy functions) is indicated in gray.

Used abbreviations:

FP = failure probability,
 LSF = limit state function,
 FOSM = First Order Second Moment
 FORM = First Order Reliability Method
 MC=Monte Carlo,
 DS=Directional Sampling

1. Import or edit basic LSF input

Selected stress model, see section 4
 Name of the input file for a selected external stress model
 Selected LSF types, with specific input, see section 3
 Table with all stochastic stress model parameters, see section 3 and 4
 (label, mean, standard deviation, distribution type) or (P10, P90, distribution type)

2. Edit configuration input, in case of modified defaults

Name labels of stochastic parameters
 Number of sections to consider in fault
 Maximum number of FORM iterations
 Number of samples for MC and DS
 Failure criteria for different LSF types (resistance values)

Working directory for input and output of stress models per scenario
(more settings are expected to be added during actual development)

3. Generate or edit scenario input tables

Generate or input selected type:
 a. for all parameters: single variations (P90) in one direction: $n+1$
 (for proxy functions and tornado diagrams in case of linear LSF's)
 b. for all parameters: single variation in two directions (P10,P90): $2n + 1$
 (for tornado diagrams in case of nonlinear models)
 c. combinations of parameter variations in two directions (P10,P90): $2^n + 1$
 (for proxy functions in case of nonlinear models)
 d. step-wise variation of a single parameter in case of nonlinear models
 e. manual definition
 Store the scenario input table in case of manual definition
 (per scenario: parameter values)

4. Perform scenario analysis

For each scenario:
 Extend input for the stress model with parameter values
 Call the stress model and retrieve basic output
 For each LSF and location
 (in case of principal stress based LSF: for all stress tensor components)
 Calculate LSF/stress tensor value, using the basic output
 Add row to scenario results table
 Store the scenario results table
 (per scenario: LSF type, LSF location, LSF/stress tensor value)

5a. Setup linear Proxy Functions for LSF's and stress components, using results from scenario analysis type a

- For each solicitation value in the results table
 - Calculate derivatives of the LSF/stress value to parameter variation
 - Add row to proxy table
- Store linear proxy table
 - (LSF/stress type, LSF/stress location, coefficients)

5b. Setup non-linear Proxy Functions, using basis results from scenario analysis type c

- For each basic output value in the results table
 - Perform (iterative) least squares fit using nonlinear regression function
 - Add row to a proxy table
- Store nonlinear proxy table
 - (LSF type, LSF location, fit error, coefficients)

6a. Perform probabilistic FOSM analysis for one selected linear LSF in one location

- Get LSF proxy function
- Setup parameter covariance matrix
- Perform FOSM procedure with linear LSF proxy function,
 - see section 2.5.2.5 and 4.1.2
- Store FOSM results table
 - (LSF type, LSF location, mean, standard deviation, FP)

6b. Perform probabilistic FORM analysis for one selected nonlinear LSF in one location

- Get stress proxy functions
- Setup parameter covariance matrix
- Perform iterative FORM procedure with LSF's using stress proxy functions,
 - see sections 2.5.2.5 and 4.1.3
- Store FORM results table
 - (LSF type, LSF location, mean, standard deviation, FP, design point (x and u values))

6c. Perform probabilistic sampling analysis for all selected LSF's in all locations

- Setup parameter covariance matrix
- Execute sampling with MC or DS
 - For all LSF's and for system:
 - calculate LSF values (with proxy function or with model)
 - Update the FP, the mean and the standard deviation
- Store sampling results table
 - (per LSF + system: mean, standard deviation, FP)

7 Export or view results

- generated scenario input table
- scenario results table (LSF/solicitation values)
- linear proxy table
- nonlinear proxy table
- Tornado diagram, for scenario tables type a and b
- Graphs of LSF/solicitation versus parameter for scenario tables type a and d
- FOSM results table
- FORM results table
- Diagram of FORM u parameter values in design point (importance factors)
- Sampling results table

5.5 Proposed design and proposed development platform

The proposed design of CORA encapsulates the following three layers:

- The User-Interface. This is the layer where input is collected and output is generated. It is envisaged that this UI will have a rudimentary graphical version (GUI) as well as a batch/console version. The GUI will be developed in C#, using an existing Deltares library for input of tables and for basic visualization in diagrams and graphs. Input, intermediate results and final results will internally be stored in and retrieve from XML files. Export facilities of the generated results tables to Excel will facility a broad range of additional external visualization options.
- The business layer. This layer contains the logic for the execution of a scenario analysis, setting up proxy functions and performing a probabilistic analysis. The business layer will also be developed in the C# language and link to an existing Deltares (FORTRAN) library for probabilistic analysis. The probabilistic library supports the FORM, Monte Carlo and Directional sampling methods and various types of distribution functions. The existing Monte Carlo and Directional sampling code will need some modification, in order to support the simultaneous output for different LSF's at local level together with the global limit state surface at system level.
- A layer for the calculation of LSF values (either by using a stress model or by proxy tables). This layer will be contained in an external DLL, which can be altered and extended independent of CORA itself. The interface between CORA and the library will be defined in terms of a generic .NET (dotNET) API (Application Protocol Interface). This interface support all DLL's developed in any language supporting the Microsoft .NET framework. See <http://dotnetpowered.com/languages.aspx> for an overview of supporting languages, including for example Visual Basic.

The proposed language for version 1.1 of the DLL is C# or VB.NET, depending on the preference of Shell.

It is proposed that version 1.1 of the DLL will contain only the Geertsma model, as this is sufficient for a full proof of concept. The following extension of the DLL version 1.2 to Quickblocks and Geomec/Diana should require no adaptation in the business layer.

The UI and business layer of CORA will be contained in one executable, consisting partially of existing Deltares proprietary software (GUI component library, probabilistic library). The DLL for the calculation of the LSF values will be delivered also as source code: this will allow Shell experts to alter and extend the DLL themselves, when desired.

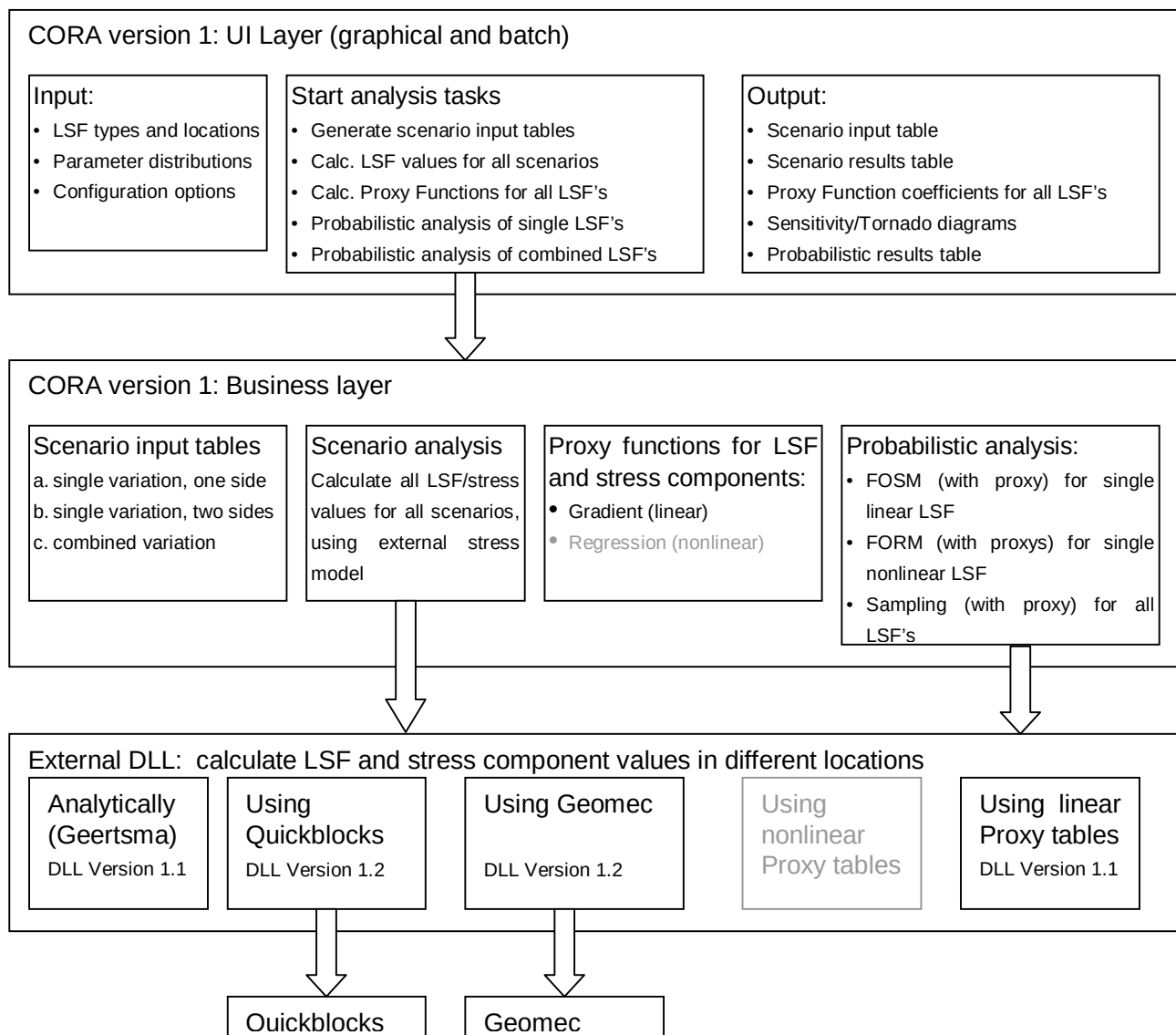


Figure 5.2: Design of CORA version 1

6 Estimates of work for version 1

6.1 Assumptions and constraints

- Two UI types will be supported: graphical and batch, with a functionality as defined in chapter 5. Early prototyping in interaction with representative Shell users will be used to define the required look and feel of the UI.
- Business layer functionality as defined in chapter 5. This layer will supply a full framework for application with different LSF definitions and different linear elastic stress models. Future extension to nonlinear stress models with intermediate nonlinear proxy functions is envisaged, but will not be part of version 1.
- The current work estimates are restricted to version 1.1 of the DLL for LSF value. This version contains only the Geertsma model. Version 1.2 will also contain connections to Quickblocks and GeoMec.
- The CORA software will partly use existing (and already tested) Deltares software components for the GUI and the probabilistic procedures.
- Quality Assurance.
 - The Deltares development method is based on continuous test integration. This means that a test suite will be set up during the code development, with coverage of the full functionality. One or more tests for each software unit will be defined prior to the development of the software unit itself.
 - The acceptance test will include the calculation of all defined different limit state function types at multiple locations.
- Version 1 will not need a users manual or a description of background.

6.2 Work Break down Structure and estimated cost range

The following table gives a rough work breakdown structure with a first estimate of the associated cost range per item. These first estimates have no formal status. Before a formal quotation, the final values will most likely be subjected to changes as a result of a refined definition of work and a refined cost analysis.

Task	Cost estimate (± 25 %) in Euros', excl. VAT
UI layer (graphical and batch) in total (incl. early prototyping)	
Geertsma DLL with 1 st LSF type	
Add 8 other LSF types to DLL	
Generate scenario input tables and execute scenario analysis	
Setup linear proxy function table from scenario results	
Execute FOSM analysis per single linear LSF	
Execute FORM analysis per single nonlinear LSF	
Execute Monte Carlo and Direction Sampling system analysis	
System acceptance tests + bug fixes	
Total	

6.3 Estimated elapse time

The required elapse time is approximately 6 months.

A Glossary of terms

OSPAR selection from a comprehensive glossary in Annex II to IPCC SRCCS.

Acid gas	Any gas mixture that turns to an acid when dissolved in water (normally refers to H ₂ S + CO ₂ from sour gas (q.v.)).
Anthropogenic source	Source that is man-made as opposed to natural.
Aquifer	Geological structure containing water and with significant permeability to allow flow.
Baseline	The datum against which change is measured.
Blow-out	Refers to catastrophic failure of a well when the petroleum fluids or water flow unrestricted to the surface.
Brine	Water with a high concentration of dissolved salts.
Buoyancy	Tendency of a fluid or solid to rise through a fluid of higher density.
Cap rock	Rock of very low permeability that acts as an upper seal to prevent fluid flow out of a formation.
Capillary entry pressure	Additional pressure needed for a liquid or gas to enter a pore and overcome surface tension.
CO ₂ stream	Carbon dioxide streams from carbon dioxide capture processes for storage in geological formations, which consist overwhelmingly of carbon dioxide. They may contain incidental associated substances derived from the source material and the capture, transport and storage processes used, i.e.: - source and process derived substances; and - added substances (i.e. substances added to the CO₂ stream to enable or improve the capture, transport and storage processes).
Casing	A pipe, which is inserted to stabilise the borehole of a well, after it is drilled.
CO ₂ capture and storage	This is a process consisting of the separation of a CO ₂ stream from industrial and energy-related sources, transport to a storage location and long-term isolation from the biosphere, including the atmosphere.
Completion of a well	Refers to the cementing and perforating of casing and stimulation to connect a well bore to a formation.
Confinement	The process by which a CO ₂ stream is kept within a specified geological space.
Containment	Restriction of movement of a fluid to a designated volume (e.g., a reservoir).
D, Darcy	A non-SI unit of permeability, abbreviated D, and

	approximately = 1µm ² .
Deep saline aquifer	A deep underground rock formation composed of permeable materials and containing highly saline fluids.
Dense fluid	A gas compressed to a density approaching that of the liquid.
Dense phase	A gas compressed to a density approaching that of the liquid.
Depletion	Of a reservoir: where production is significantly reduced.
Dissolution	With respect to CO ₂ , the process by which CO ₂ separates into its component ions in water.
EOR	Enhanced oil recovery: the recovery of oil additional to that produced by standard production methods.
Fault	In geology, a surface at which strata are not longer continuous but displaced.
Flood	The injection of a fluid into an underground formation.
Formation	A body of rock of considerable extent with distinctive characteristics that allow geologists to map, describe, and characterise it.
Formation water	Water that occurs naturally within the pores of rock formations.
Fracture	Any break in rock along which no significant movement has occurred, but where the permeability may be significantly enhanced.
Geochemical trapping	The retention of injected CO ₂ by geochemical reactions.
Geological time	The time over which geological processes take place.
Geomechanics	The process of movement or potential movement of rocks within the Earth's crust.
Geosphere	The Earth, its rocks and minerals, and its ground waters.
GHG	Greenhouse gases: carbon dioxide (CO ₂), methane (CH ₄), nitrous oxide (N ₂ O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulphur hexafluoride (SF ₆).
Hazard	The potential to cause harm.
Hydro-geological	Concerning water in the geological environment.
Hydrostatic	Pertaining to the properties of a stationary body of water.
IEA GHG	International Energy Agency – Greenhouse Gas R&D Programme.
Igneous	Rock formed when molten rock (magma) has cooled and solidified (crystallised).

Injection well	A well in which fluids are injected rather than produced.
Injectivity	A measure of the rate at which a quantity of fluid can be injected into a geological formation.
In-situ mineralisation	A process whereby carbon dioxide injected into a geological formation reacts with silicate minerals, forming stable carbonate minerals.
IPCC	Intergovernmental Panel on Climate Change.
Leakage	In respect of storage of CO ₂ streams, the escape of that CO ₂ stream from the storage formation into overlying formations, the water column and the atmosphere.
Log	Records taken during or after the drilling of a well.
Long term	The term following the closure of the CO ₂ storage site. This could extend to several thousand years into the future.
Mature sedimentary basins	Geological basins formed by the deposition of sedimentary particles and grains under sub-aqueous and sub-aerial conditions and in which deposited organic matter has matured into hydrocarbon reserves.
Microseismicity	Small-scale seismic activity, usually only detectable by the use of sensitive instrumentation.
Migration	The movement of fluids within or out of formations.
Mitigation	The process of reducing the adverse impact of any failure in the CO ₂ storage system.
Monte Carlo simulation	A modelling technique in which the statistical properties of outcomes are tested by random inputs.
Mudstone	A very fine-grained sedimentary rock that commonly provides a seal, thus preventing the upward migration of fluids.
Observation well	A well installed to permit the direct observation of subsurface conditions.
OSPAR Convention	Convention for the Protection of the Marine Environment of the North-East Atlantic, adopted in Paris, 22 September 1992.
Other substances (or associated substances)	Associated substances originating from the source material and the capture, transport and storage processes used.
Overburden	Rocks and sediments above any particular stratum.
Overpressure	Pressure created in a formation that exceeds the pressure inherent at the formation's depth.
Permanence	The term to indicate the likelihood that the situation will stay unchanged.
Permeability	Ability to flow or transmit fluids through a porous solid such as rock.

Pore space	Space between sedimentary grains that can contain fluids.
Porosity	Measure of the amount of pore space in a rock.
Regional scale	A geological feature that crosses an entire basin, or other geological provinces.
Remediation	The process of correcting any source of failure, for example in a CO ₂ storage system.
Reservoir	A subsurface body of rock with sufficient porosity and permeability to store and transmit fluids.
Risk	Probability of occurrence of an undesired event, multiplied by the (HSE) impact of that event.
Risk assessment	Part of a risk-management system, consisting of exposure assessment, effect assessment and risk characterisation.
Risk characterisation	Risk characterisation is the step in the risk assessment process which determines the likelihood and severity of impacts on the marine environment.
Saline formation	Sediment or other rock formation containing brackish water or brine.
Seal	An impermeable rock that forms a barrier above or around a formation such that fluids are held in the formation.
Seismic technique	Measurement of the properties of rocks by the refraction and reflection of sound waves generated artificially or naturally.
Shale	Impermeable very fine-grained and finely laminated sediment that commonly provides a seal to the movement of underlying fluids.
Short term	The near term prior to closure of the CO ₂ storage site. This could extend to some one hundred years into the future.
Sour gas	Natural gas containing significant quantities of acid gases, such as H ₂ S and CO ₂ .
Spill point	The structurally lowest point in a structural trap (q.v.) that can retain fluids lighter than background fluids.
Storage	A process for retaining captured CO ₂ streams in deep geological formations so that it does not reach the atmosphere. The terms sequestration and storage are also used interchangeably.
Storage site	The location for storage in geological formations, comprising one or more wellheads and surface facilities.
Structure	Geological feature produced by the deformation of the Earth's crust, such as a fold or a fault; a feature within a rock such as a fracture; or, more generally, the

	spatial arrangement of rocks.
Supercritical	At a temperature and pressure above the critical temperature and pressure of the substance concerned. The critical point represents the highest temperature and pressure at which the substance can exist as a vapour and a liquid in equilibrium.
Tectonically active area	Area of the Earth where deformation is presently causing structural changes.
Trap	A geological structure that physically retains fluids, which are lighter than the background fluids.
Well	Manmade hole drilled into the Earth to produce liquids or gases, to allow the injection of fluids, or to enable observations of subsurface process.
Well integrity	The ability of a well to prevent any leaks from occurring, either along the (cemented) annulus (casing / open hole) or between the plugs and the casing.
Wellhead pressure	Pressure developed at the top of the well.

